

Prompt-Conditioned Naturalness Distributions for Speech-to-Speech Evaluation

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Abstract—Speech-to-speech (S2S) systems are increasingly capable of producing fluent spoken responses, but their evaluation remains weakly tied to the distributional structure of natural conversation. We propose a prompt-conditioned evaluation framework that learns the expected distribution of response-side speech metrics from natural dyadic interactions. Given prompt-side acoustic, temporal, affective, speaker-profile, conversational-context, and speech-embedding features, the model predicts a distribution over target response metrics including duration, lexical length, prosody, rhythm, pausing, affect, and speaker-profile estimates. An S2S output can then be scored by extracting the same response metrics and comparing them against the learned conditional distribution for the corresponding prompt. The framework supports deterministic baselines, diagonal Gaussian predictors, conditional variational autoencoders, and embedding-conditioned variants using Whisper and Trillsson prompt representations. This paper describes the dataset construction, metric space, modeling formulation, and planned uncertainty-aware scoring procedure for interpretable S2S naturalness evaluation.

Index Terms—speech-to-speech evaluation, conversational naturalness, conditional generative modeling, speech embeddings, variational autoencoders

I. INTRODUCTION

Spoken dialogue is shaped by more than lexical content. Natural responses occupy context-dependent ranges of duration, speaking rate, pausing, pitch behavior, affective tone, and speaker-state cues. As S2S systems move from read-speech synthesis toward interactive spoken agents, evaluation methods must capture whether an output is plausible for the local conversational context rather than merely intelligible or textually correct.

Many current evaluation practices for spoken dialogue systems remain centered on automatic speech recognition, task completion, text similarity, or global human ratings. These signals are useful, but they do not directly explain why a generated response sounds natural or unnatural as speech. For example, an S2S response may be semantically appropriate but too slow for a high-arousal prompt, too long for a short backchannel context, too monotone for an expressive exchange, or temporally mismatched to the turn transition. Such failures are naturally described as deviations from conditional distributions over speech behavior.

Our prior analysis of the Seamless Interaction dataset established large-scale reference regimes for conversational prosody and rhythm, showing that natural operating ranges shift with speaker and interaction factors. The present work turns that descriptive finding into a predictive evaluator. Rather than

defining one pooled acceptable range for each metric, we learn a prompt-conditioned distribution over response-side metrics. At evaluation time, an S2S waveform is mapped into the same metric space and scored by its likelihood, calibrated interval membership, and per-feature standardized residuals under the predicted distribution.

The central object is a conditional model

$$p_{\theta}(y \mid x), \quad (1)$$

where x contains prompt-side acoustic, temporal, affective, speaker-profile, context, and embedding features, and y contains response-side target metrics. The model conditions only on the prompt and available interaction context. The S2S response itself is not used as conditioning information; it is the object being scored.

This paper contributes a distributional formulation for S2S naturalness evaluation, a paired prompt-response metric pipeline over natural conversational speech, and a modeling framework that compares deterministic, Gaussian, conditional variational autoencoder (CVAE), and embedding-conditioned predictors. The final experimental results are in progress; accordingly, this draft focuses on the completed methodological design and leaves the Results and Conclusion sections as placeholders.

II. RELATED WORK

A. Speech-Native S2S Evaluation

S2S evaluation has often inherited methods from automatic speech recognition, text generation, and text-to-speech evaluation. These approaches measure important aspects of system quality, but they can underrepresent conversational timing, turn-taking fit, prosodic expressivity, and context-appropriate response behavior. Subjective ratings remain essential, yet they are expensive to collect and often provide limited diagnostic detail. The goal of the present work is not to replace human evaluation, but to provide a scalable speech-native diagnostic that can identify which measurable dimensions of a response deviate from natural conversational behavior.

B. Conversational Prosody and Rhythm

Natural conversation is coordinated through timing and prosodic cues that help manage turns, signal stance, and shape perceived engagement. Prior work has shown that conversational speech differs from read speech in both prosodic and temporal structure, and that speaking rate, pausing, and pitch

variation shift with speaker traits and interactional state. Our preceding Seamless Interaction study used over 4,000 hours of dyadic English conversation to estimate empirical operating regimes for prosodic and temporal metrics. Those regimes motivate the present conditional formulation: naturalness should be evaluated relative to a matched prompt and context, not against a single global baseline.

C. Distributional Modeling

Point predictions are insufficient when the target object is a range of plausible responses. For S2S evaluation, multiple different response patterns may be natural for the same prompt, and a model should represent this uncertainty. We therefore compare mean-only predictors with probabilistic models that produce conditional distributions. Gaussian neural predictors directly estimate a diagonal normal distribution over metrics, while CVAEs use latent variables to represent response variability conditioned on prompt context [1], [2].

D. Speech Embeddings

Hand-engineered acoustic and temporal metrics summarize interpretable dimensions of speech, but they may miss broader information about prompt content, speaking style, and paralinguistic context. We therefore incorporate learned prompt representations from two complementary embedding families. Whisper hidden-state embeddings provide representations from a large-scale weakly supervised speech recognition model [3]. Trillsson embeddings provide non-semantic speech representations designed for paralinguistic information [4]. In the proposed evaluator, these embeddings augment the prompt-side conditioning vector rather than replacing interpretable metric features.

III. DATASET AND METRICS

A. Paired Prompt-Response Construction

We construct paired examples from the Seamless Interaction dataset, a large corpus of dyadic English conversations containing naturalistic and improvised interactions. Each training example corresponds to an adjacent prompt-response pair. The prompt segment provides conditioning information, and the following response segment provides the target metric vector.

Each pair contains identifiers for the session, interaction, dyad, prompt speaker, response speaker, and turn index; timing fields for prompt and response boundaries; response latency and overlap; transcripts; relationship metadata; and the extracted metric fields. Identity-like columns are used for bookkeeping and split construction but are not intended as direct predictive features in the final evaluator, in order to reduce leakage from speaker or session identity. The final train, validation, and test split policy is still being finalized and should be reported here once experiments are locked.

B. Prompt Conditioning Features

The conditioning vector x contains prompt-side speech measurements and available conversational context. The hand-engineered prompt metrics include duration, word count, fundamental frequency (F_0) mean, F_0 standard deviation, F_0

range, voiced ratio, local speech rate, local articulation rate, pause ratio, mean pause duration, arousal, valence, dominance, estimated age, predicted gender information, gender confidence, Vox-Profile chunk count, and support speech duration. Contextual fields include turn index, response latency or overlap information available from the pair construction, relationship metadata, and speaker-role fields. Numeric fields are standardized after imputation, and categorical fields are encoded for neural modeling.

C. Response Target Metrics

The target vector $y \in \mathbb{R}^{17}$ contains response-side metrics chosen to describe conversational naturalness across lexical, prosodic, temporal, affective, and speaker-profile dimensions. The 17 metrics are response duration, word count, F_0 mean, F_0 standard deviation, F_0 range, voiced ratio, speech rate, articulation rate, pause ratio, mean pause duration, arousal, valence, dominance, estimated age, gender confidence, Vox-Profile chunk count, and Vox-Profile support speech seconds. Prosodic features are computed from voiced regions, temporal features are derived from word timing and pause structure, and affective and speaker-profile estimates are produced with Vox-Profile-style speech trait models. These metrics match the dimensions used to score future S2S outputs, making the evaluator directly interpretable at the feature level.

D. Vox-Profile Annotations

The dataset does not provide all speaker and interaction-state annotations needed for conditional naturalness modeling. We therefore augment the pair table with Vox-Profile estimates for age, gender-related posterior information, and dimensional affect. The age and gender outputs are treated as model-derived speech profile estimates rather than ground-truth demographics. Arousal, valence, and dominance are used as continuous descriptors of interactional state. Because these annotations are predicted rather than self-reported, final interpretation should distinguish model-conditioned stratification from claims about true demographic attributes.

IV. MODELING APPROACH

A. Problem Formulation

For each prompt-response pair i , the model observes a prompt-side conditioning vector x_i and a response-side target vector y_i . The learning objective is to estimate the conditional distribution in (1). For an S2S system output, the same response-side metrics are extracted to form $\hat{y}_{s2s}$, which is then compared against $p_\theta(y | x)$ for the original prompt.

This setup separates generation from evaluation. The model does not attempt to synthesize speech or text. It learns what response metric distributions are plausible in natural human dialogue for a given prompt and context. The resulting evaluator can provide both a scalar plausibility score and per-metric diagnostics.

B. Deterministic Baseline

The deterministic baseline predicts the conditional mean:

$$\hat{y} = f_{\theta}(x). \quad (2)$$

The model is trained with mean squared error in standardized target space. This baseline tests how much of the response metric structure is predictable without explicitly modeling uncertainty. It also provides a simple reference for original-scale error analysis.

C. Gaussian Predictor

The Gaussian predictor estimates a diagonal normal distribution:

$$p_{\theta}(y | x) = \mathcal{N}(y; \mu_{\theta}(x), \text{diag}(\sigma_{\theta}^2(x))). \quad (3)$$

The network has separate heads for the conditional mean and log variance, and it is trained using Gaussian negative log-likelihood. This model supports likelihood-based scoring, interval estimates, and per-feature uncertainty diagnostics while remaining simpler than latent-variable models.

D. Conditional Variational Autoencoder

The CVAE introduces a latent variable z to represent conditional response variability. The encoder approximates $q_{\phi}(z | x, y)$, and the decoder predicts a diagonal Gaussian distribution over y from (x, z) . Training minimizes

$$\mathcal{L} = -\log p_{\theta}(y | x, z) + \beta D_{\text{KL}}(q_{\phi}(z | x, y) \| p(z)), \quad (4)$$

where $p(z)$ is a standard normal prior and β controls the strength of the KL regularization. At evaluation time, samples from $p(z)$ yield Monte Carlo estimates of the predictive distribution. This formulation is intended to capture one-to-many response structure: several response metric configurations may be natural for the same prompt.

V. EMBEDDING CONDITIONING

A. Whisper Prompt Embeddings

The embedding-enabled pipeline merges pair-level Whisper representations with the tabular prompt features. For each prompt segment, a 1,280-dimensional vector is loaded from the embedding table and concatenated with the processed conditioning representation. These vectors are derived from hidden states of a Whisper-based speech model and are intended to encode broader information about the prompt signal than the hand-engineered metrics alone.

B. Trillsson Prompt Embeddings

We also extract Trillsson4 embeddings for the paired prompt and response segments using the TensorFlow Hub Trillsson model. For evaluator conditioning, the prompt-side Trillsson vector is used as an additional representation of the prompt. The current extraction pipeline saves pooled 1,024-dimensional Trillsson vectors at 16 kHz for both prompt and response segments, with the prompt vector serving as the model input stream for $p_{\theta}(y | x)$. Trillsson is included because non-semantic speech representations are designed to preserve

paralinguistic information such as speaker state, style, and acoustic context.

C. Fusion Variants

Embedding-conditioned models test whether learned speech representations improve conditional prediction beyond interpretable tabular features. The planned fusion variants include simple concatenation after projection, late fusion with separate tabular and embedding encoders, stronger learned fusion layers, and cross-encoder-style fusion. The paper will report results only after all variants are evaluated under a common split, preprocessing pipeline, target scaling strategy, and metric suite.

VI. SCORING FRAMEWORK

A. Likelihood and Residual Scores

Given a prompt x and an S2S response metric vector $\hat{y}_{\text{s2s}}$, probabilistic models can assign a negative log-likelihood score:

$$\text{NLL}(\hat{y}_{\text{s2s}}, x) = -\log p_{\theta}(\hat{y}_{\text{s2s}} | x). \quad (5)$$

Lower NLL indicates that the output lies in a higher-density region of the learned natural-response distribution. For model families that output per-feature variance estimates, we also compute standardized residuals:

$$r_j = \frac{\hat{y}_{\text{s2s},j} - \mu_j(x)}{\sigma_j(x)}. \quad (6)$$

These residuals identify the direction and magnitude of each deviation. For example, an output may be globally plausible but have an unusually long duration, compressed pitch range, or atypical pause ratio for its prompt.

B. Interval and Calibration Diagnostics

The evaluator should also report whether each observed metric lies inside model-predicted intervals. For Gaussian predictors, intervals can be obtained directly from $\mu_j(x)$ and $\sigma_j(x)$. For CVAE models, intervals can be estimated from Monte Carlo predictive samples. Coverage at nominal levels such as 50%, 80%, 90%, and 95% provides a calibration diagnostic: a useful evaluator should not only predict accurate means, but also assign realistic uncertainty to natural response variability.

C. Interpretable Aggregation

A practical S2S diagnostic can aggregate likelihood, residual, and interval signals into a scalar naturalness score while preserving feature-level explanations. The scalar score supports system comparison, while the per-feature diagnostics support development and error analysis. This is important for S2S systems because unnaturalness is often localized: a response may have appropriate content and affect but fail through timing, pausing, or prosodic range.

VII. EXPERIMENTAL PROTOCOL

The final experiments will compare deterministic MLP, Gaussian MLP, CVAE, and embedding-conditioned CVAE variants. All models will be evaluated under the same split policy, preprocessing state, target scaling, and response-metric extraction procedure. The finalized split construction is still TBD and will be specified before reporting results.

The planned evaluation metrics include standardized-space prediction error, original-scale per-target RMSE and MAE, probabilistic negative log-likelihood, interval coverage, and calibration summaries. Embedding models will compare tabular-only conditioning against Whisper-augmented, Trillsson-augmented, and combined embedding conditions where appropriate. No numerical results are reported in this draft because experiments are still in progress.

VIII. RESULTS

IX. CONCLUSION

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